



Project 5

Design and Simulation of MIPS Register File in VHDL

1. In this project, you will build and simulate the operation of the MIPS register file. As we discussed in class, the register file is one of the subsystems inside the processor. It contains all 32 integer registers, each 32 bits wide. Below is a block diagram of the MIPS register file along with all the input/output signals which we discussed and used during our classes. However, the descriptions of these signals are provided below to help define the signals in your VHDL code.



Figure 1: Block diagram of MIPS register file with all input/output signals.

All input/output signals to/from the register file are explained below:

Input Signals:

- **Read Register 1:** A 5-bit signal that selects one of the 32 registers to be read; its data contents will be placed at the “Read Data 1” output of the register file. Name this signal **ReadReg1** in your VHDL code.
- **Read Register 2:** A 5-bit signal that selects one of the 32 registers to be read; its data contents will be placed at the “Read Data 2” output of the register file. Name this signal **ReadReg2** in your VHDL code.
- **Write Register:** A 5-bit signal that selects one of the 32 registers into which data will be written, as defined by the “Write Data” input. Name this signal **WriteReg** in your VHDL code.

- **Write Data:** A 32-bit signal which is the data that we want to write into the register defined by the “Write Register” input of the register file. Name this signal **WriteData** in your VHDL code.
- **RegWrite:** A 1-bit control signal to activate the write operation to a register. Name this signal **RegWrite** in your VHDL code.
- **Clk:** 1-bit clock signal which we will add to synchronize all operations of the register file to the positive edge of the clock. Name this signal **Clk** in your VHDL code.

Output Signals:

- **Read Data 1:** A 32-bit signal representing the content of the register defined at “Read Register 1”. Name this signal **ReadData 1** in your VHDL code.
 - **Read Data2:** A 32-bit signal representing the content of the register defined at “Read Register 2”. Name this signal **ReadData2** in your VHDL code.
2. Create a new project and name it **Project5**. Write the VHDL description of the MIPS register file as explained above. Name your VHDL file as **RegFile.vhd**
 3. Use the simulation to verify the correctness of your code. Please use the following values in your simulation.

Registers t0, t1, t2, t3, and t4 have the following initial values (in hex format) BA1234A5, 3456BACC, ACF33542, FFFF1D00, and 12345678, respectively. All other registers have 0 initial value.

Read register t0 at time = 1 ns

Read register t3 at time = 2 ns

Write $(A1B2C3D4)_{hex}$ into register t5 at time = 2 ns

Write Data: You need to set it correctly.

Clk period is 1 ns.

Submissions:

1. Screenshot of the simulation result window
2. Your RegFile.vdd file